Disaster Tweet Classification Using Natural Language Processing (NLP) techniques and Deep Learning models

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***Abstract*—Disaster-related information shared on social media is an important tool in the response and management of disasters, but it is also difficult to differentiate actual disaster and non-disaster-related information. This issue is solved through this project, which was conducted by means of binary classification of disaster-related tweets with the help of Natural Language Processing and machine learning. The dataset was composed of real-world tweets as disaster and non-disaster and preprocessed with the help of text cleaning, text tokenization, and text vectorization. A baseline TF-IDF with Logistic Regression model was run and compared to a number of deep learning models, such as CNN with pretrained GloVe embeddings, BiLSTM, and BiGRU, to help identify both contextual and sequential text patterns. Moreover, a transformer based model was used to utilize self-attention in modeling long range dependencies. Experimental results show that deep learning and transformer-based models achieve better performance compared to traditional machine learning approaches. The results reveal that a combination of pre-trained embeddings and sophisticated neural structures is effective in disaster tweet classification.**

***Index Terms*—Disaster Tweet Classification, Natural Language Processing, Machine Learning, TF-IDF, GloVe Embeddings, CNN, BiLSTM, BiGRU, Transformer Models.**

I. Introduction

Social media platforms such as Twitter have become vital sources of real-time information during natural disasters, accidents, and emergencies. Individuals often post updates, warnings, and requests for help, which can support disaster response agencies in making timely decisions. However, social media content is highly unstructured, noisy, and filled with abbreviations, slang, emojis, and misspellings, making automatic analysis difficult.

Manually filtering disaster-related tweets is impractical due to the massive volume of social media data generated during emergencies. Automated tweet classification systems can help identify critical information quickly and accurately. While traditional machine learning methods offer simplicity and interpretability, they struggle with contextual understanding. Recent advances in deep learning and transformer-based models provide promising solutions by capturing semantic and contextual information more effectively.

During disasters, people post important information on Twitter, but the tweets are often short, noisy and written in informal language. Traditional machine-learning models struggle to correctly identify which tweets are actually related to a disaster because they depend on simple features like keywords, bag-of-words, or TF-IDF, which cannot capture context or slang. Modern deep-learning and transformer-based methods show much better performance, but there is still no single clear approach that can consistently handle problems like lack of labeled data, domain shifts between disasters, and the highly variable writing style of users. Therefore, the main problem is to analyze and compare different machine-learning, deep-learning, and embedding techniques to accurately classify disaster-related tweets, while addressing challenges such as noisy text, limited labels, and poor generalization across different disaster events [1].

II. Literature Review

There has been considerable research on predicting disaster-related information from social media by analyzing text patterns, contextual hints, and historical tweet data. The initial works mostly used standard machine learning tools and hand-crafted features. Imran et al. published one of the initial large-scale human-annotated crisis Twitter datasets, containing more than 52 million Twitter messages of 19 disaster events, and suggested that Word2Vec embeddings and normalization are significant tools for future research [1]. Sakaki et al. demonstrated that real-time Twitter stream monitoring can detect and trace the location of disaster events, positioning social media as a crowdsourced emergency sensor [2]. Subsequent studies highlighted that shallow keyword-based models struggle with informal, short, and highly contextual tweets. Sarkar and Gonzalez-Hernandez surveyed the development of rule-based and TF-IDF approaches in deep learning and transformer models, noting challenges such as slang, multilingual text, data bias, fake news, and poor disaster generalization [3].

Later research addressed these issues using word embeddings and deep neural networks. Algiriyage et al. found that deep learning models, especially BiLSTM and CNN, with pre-trained embeddings like Word2Vec, FastText, and GloVe,

achieve high F1-scores in 46 disaster datasets, and BiLSTM better captures context [4]. Deb and Chanda showed that BERT-based contextual embeddings achieve high accuracy in disaster tweet classification [5]. Transformer models, particularly BERT, consistently outperform classical TF-IDF-based classifiers, as shown by Le and by Bousbib and Xavier [6], [7]. Nguyen et al. found that including domain knowledge, disaster-specific keywords, and tweet metadata improves CNN and LSTM models in noisy settings [8]. Li et al. proposed domain-adaptive models allowing prediction for new disasters with minimal labeled data [9]. Recent works like CrisisMatch focus on few-shot and semi-supervised learning to handle early disaster phases with limited labels [10].

Overall, the literature shows a shift from traditional ML to deep and transformer-based architectures for disaster tweet classification, encouraging hybrid modeling approaches combining classical, deep sequential, and transformer methods.

III. Dataset Description

The dataset of this project comprises tweets on Twitter and manually categorized as a disaster-related problem or not, i.e. non-disaster-related, which is binary text classification. The dataset has an identifier (id) and the content of the tweet (text) and a target label (target) where 1 represents a disaster and 0 represents a non-disaster tweet. There are no instances of missing values in these main columns of the dataset and hence there is consistency in the data used to train and evaluate. Since Twitter text is informal and noisy, preprocessing and feature extraction is essential to the attainment of reliable classification performance.

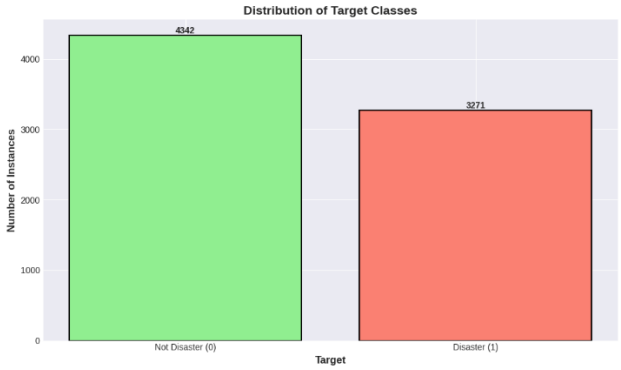

Fig. 1. Distribution of disaster vs non-disaster tweets

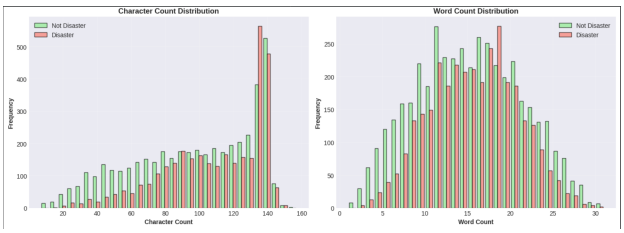

Fig. 2. character and word count distribution

These histograms in 2 compare character and word counts across both classes, revealing that both disaster and non-disaster tweets follow a similar distribution pattern, typically peaking near the 140-character limit.

A. Statistics of Tweets and vocabulary

A review of the dataset shows that the vocabulary is quite heterogeneous and sparse as is characteristic of social media data [12]. There are 27,983 unique words in the dataset of which 7,969 words are repeated. Besides, there are also 122 distinct characters, which indicate the usage of punctuations, symbols, and special characters, which are typical of tweets. The longest tweet in the data dataset has 157 characters, whereas the longest tweet in the training has 149 characters. Twitter is limited to 31 words in a tweet (Word of character), the same size in the entire dataset as well as in the training set. These statistics inspired the selection of fixed-length padding and models that are used in short-text classification. Besides the simple statistics of the dataset, an Exploratory Data Analysis (EDA) was also taken to gain a deeper insight into the distribution of classes, the features of the tweet length, the use of vocabulary, and the Twitter specifics of a hashtag, mention, URL and punctuation marks. These experiments were valuable in terms of understanding the type of data and selecting the appropriate models based on short and noisy social media text classification.

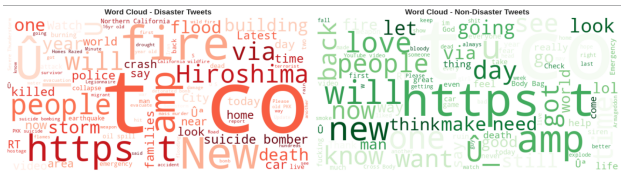

Fig. 3. Word Cloud

The word clouds in 3 highlight the most frequent terms unique to each category, where ”Disaster” tweets are dominated by urgent terms like fire, Hiroshima, and storm, while ”Non-Disaster” tweets feature more casual vocabulary like love, new, and want.

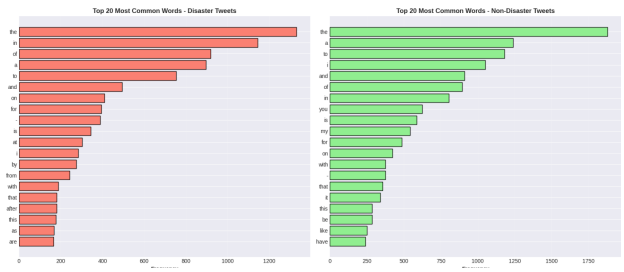

Fig. 4. Top 20 most common words

IV. Methodology

A. Common Preprocessing Pipeline

To guarantee uniformity and equitable comparison, a common preprocessing pipeline is applied to all deep learning

models used in this study (BiLSTM, BiGRU, and TextCNN). In order to sanitize raw tweet text, it is first converted to lowercase, URLs and numeric characters are removed, repeated characters are normalized, and excess whitespace is trimmed [13]. A Keras tokenizer fitted on the training set tokenized text to the top 5,000 words [14].

To create consistent input dimensions, each tokenized tweet is transformed into an integer sequence and padded to a predetermined length of 31 tokens. The embedding layer is initialized using pretrained GloVe 300-dimensional word embeddings, which offer semantically significant word representations acquired from a sizable external corpus. The layer of embedding is retained. To maintain this pretrained knowledge, the embedding layer is frozen during training. The architectural designs of the models diverge after this.

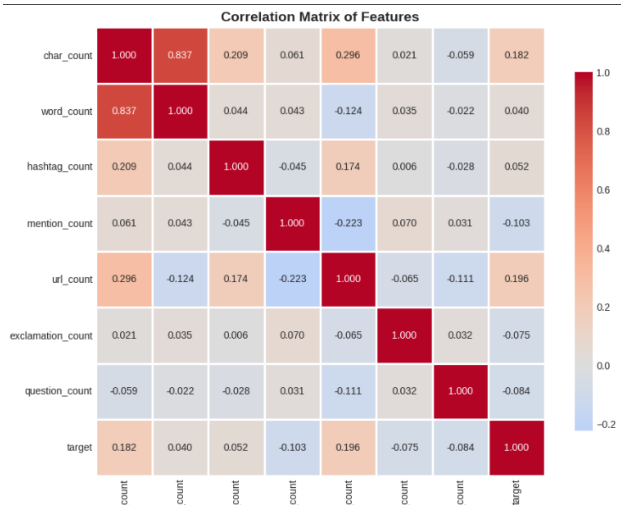

Fig. 5. Correlation Matrix of Numerical Features

The correlation matrix in **Figure 5** illustrates the relationships between the extracted numerical features. A significant positive correlation of 0.837 is observed between *char_count* and *word_count*, which is expected as longer text naturally contains more words; however, most features show a weak correlation with the *target* variable, suggesting that simple count-based features alone may not be sufficient for highly accurate classification.

B. Models and Tools

TextCNN TextCNN captures local n-gram features using 1D convolution over embeddings, followed by global max-pooling and fully connected classification [12].

Bidirectional GRU (BiGRU) BiGRU processes sequences forward and backward using gated recurrent units, capturing contextual information efficiently [15].

Bidirectional LSTM (BiLSTM) BiLSTM uses LSTM units to capture long-range dependencies bidirectionally [13].

TF-IDF + Logistic Regression TF-IDF features represent tweets as weighted n-grams, classified with Logistic Regression [12].

Transformer-Based Models Models using self-attention, such as DistilBERT, ALBERT, and MiniLM, capture global contextual relationships in parallel [14], [16], [17].

Hybrid Models

Sentence-Transformer + Logistic Regression: Generates semantic embeddings using MiniLM and classifies with Logistic Regression [18].

BiLSTM + Attention + Frozen MiniLM: MiniLM embeddings are fed to BiLSTM with attention to highlight disaster-relevant tokens.

V. Results and Evaluation

1) Baseline Model Performance (TF-IDF + Logistic Regression): The traditional machine learning method was involved to classify the disaster tweets by using the TF-IDF with Logistic Regression model and set it as a baseline to evaluate the performance. Its model also had a fair level of accuracy and F1-score, which revealed that word-frequency features are suitable in determining explicit keywords of a disaster. Nevertheless, its low recall indicates that it is not easy to obtain contexts, slang, abbreviations, and metaphorical phrases. Therefore, there were frequent cases of misclassification of context-dependent tweets. In spite of its swiftness and interpretability, the general performance of this model was evidently lower than that of deep learning and transformer-based solutions.

A. Deep Learning Model Comparison

TextCNN was the only deep learning model that outperformed the baseline with the ability to capture local n-gram features and short disaster-related phrases, which resulted in an F1-score of 0.75. Nevertheless, it could not model long-range dependencies, which made the gains in performance less. BiGRU showed superior contextual learning in a bidirectional processing and balanced precision and recall of 0.75 with a F1-score, whilst being computationally efficient. BiLSTM was the most successful amongst the non-transformer deep learning models with greater precision and recall and greater contextual information. This was especially useful because of its capability to capture long-term dependencies and therefore process ambiguous and informal text on social media.

As shown in **Figure 6**, the LSTM model achieves a high training accuracy exceeding 97%. However, the validation accuracy remains stagnant between 78% and 80%. This divergence between the two curves is a classic indicator of *overfitting*, suggesting that the model is memorizing the training noise rather than learning generalizable patterns.

Similarly, the GRU model's performance is shown in **Figure 7**. The GRU architecture reaches convergence faster than the LSTM, but it exhibits comparable results with a validation accuracy peaking at approximately 81%. Like the LSTM, the persistent gap between training and validation metrics suggests that further regularization techniques, such as Dropout or Early Stopping, might be required to improve generalization.

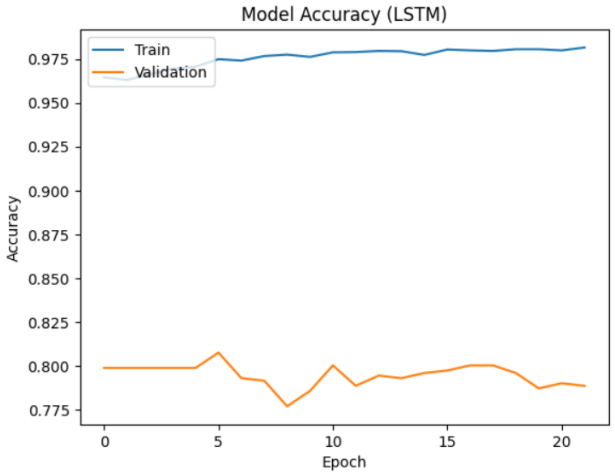

Fig. 6. Training and Validation Accuracy for the LSTM Model

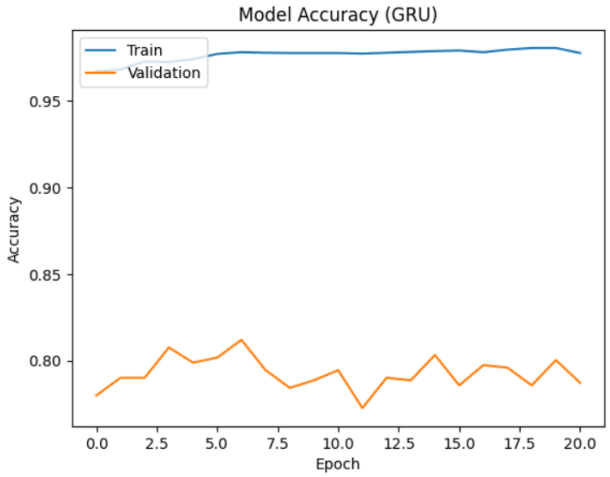

Fig. 7. Training and Validation Accuracy for the GRU Model

B. Performance of Transformer-Based Models

The best overall performance in this research was made by transformer-based models. DistilBERT gave good performance at lower computational cost whereas ALBERT gave good results with few parameters. MiniLM conformed to the optimal point of accuracy, recollection, and F1-score (0.80), which is why it is best applied to disaster monitoring in large-scale or real-time. In general, these models proved useful especially in interpreting informal language, slang, abbreviations, and sense based on context, owing to their self-attention abilities.

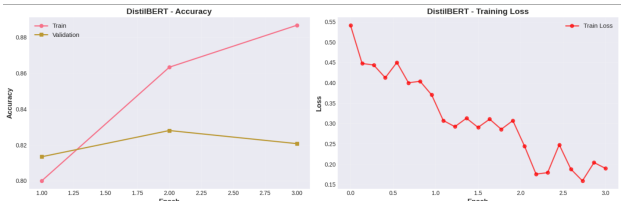

Fig. 8. DistilBERT Training Accuracy and Loss Curves.

As shown in **Figure 8**, DistilBERT was fine-tuned over 3 epochs. The training accuracy improved significantly, reaching nearly 89%, while the validation accuracy peaked at approximately 83%. The rapid decrease in training loss indicates that the pre-trained weights provide a robust starting point for disaster tweet classification.

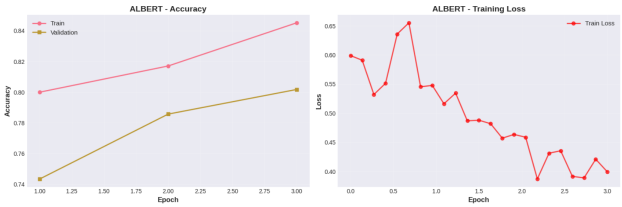

Fig. 9. ALBERT Training/Validation Accuracy and Training Loss.

The performance of the ALBERT model is depicted in **Figure 9**. Over three training epochs, the model shows a consistent increase in accuracy, with validation accuracy reaching approximately 80.2%. While the training loss displays minor fluctuations, the overall downward trend suggests effective gradient descent. ALBERT's parameter-sharing architecture allows it to achieve these competitive results with significantly fewer parameters than traditional BERT models, making it an efficient choice for this classification task.

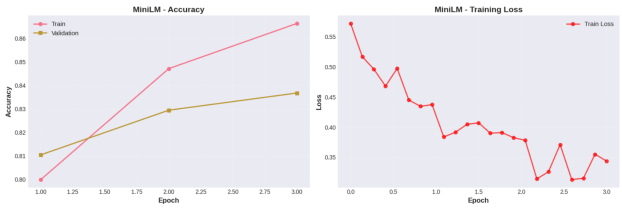

Fig. 10. MiniLM Training Accuracy and Training Loss over 3 Epochs.

The performance of the MiniLM model is presented in **Figure 10**. Over the course of 3 epochs, the training accuracy improved from 80% to over 86%, while the validation accuracy maintained a steady rise, reaching approximately 83.7%. The training loss curve indicates a successful convergence, further proving that compact transformer models like MiniLM can achieve competitive results with lower computational overhead compared to larger architectures.

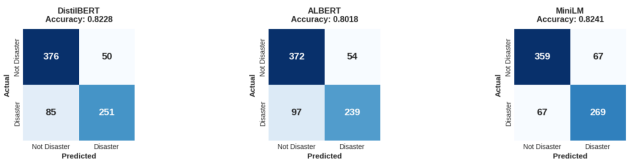

Fig. 11. Confusion matrix of DistilBERT, ALBERT, MiniLM

C. Hybrid Models

Sentence-Transformer (MiniLM) + Logistic Regression model was highly effective as compared to the TF-IDF baseline, registering high F1-score without being computational

and interpretable. This proves that traditional classifiers can be significantly improved with the help of transformer-based sentence embeddings. Frozen MiniLM + BiLSTM + Attention model had one of the best overall performances. It was able to better concentrate on the important disaster-related words by combining transformer-based semantic features, sequential modeling, and attention mechanisms, leading to better accuracy and interpretability.

According to **Figure 12**, this model achieved a validation accuracy of roughly 81.5%. Notably, the loss curves for both training and validation are tightly aligned, suggesting that the "Frozen" approach acts as a strong regularizer, effectively preventing the severe overfitting observed in the standalone RNN models.

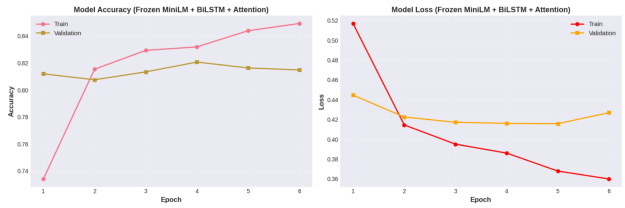

Fig. 12. Model Performance: Frozen MiniLM + BiLSTM + Attention.

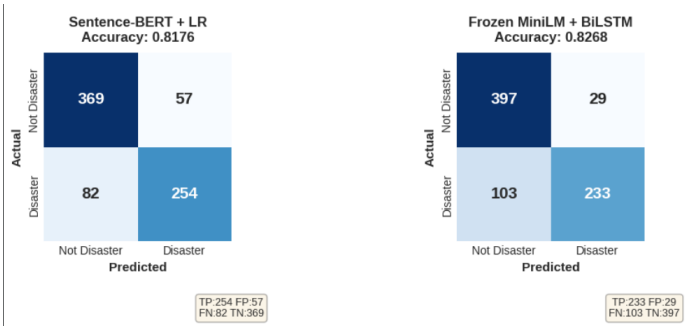

Fig. 13. Confusion matrix of Sentence-Transformer (MiniLM) + Logistic Regression and Frozen MiniLM + BiLSTM + Attention models

D. Overall Comparison Summary

In summary the findings reveal a clear trend of improved performance of context-aware models over traditional ones. TF-IDF achieved the fastest but weakest performance, while TextCNN and BiGRU achieved moderate improvements. BiLSTM became the most suitable model due to its good sequence modeling in the absence of transformers. Transformer-based and hybrid techniques provided the most favorable results, demonstrating that contextual and semantic knowledge are most important in successful disaster tweet classification.

Model performance (accuracy, precision, recall, F1-score) is shown in Table I.

VI. Discussion

Traditional ML models perform well on clear disaster keywords but struggle with context. Deep learning captures sequential relationships. Transformer and hybrid models perform best due to self-attention capturing contextual meaning.

TABLE I
Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score
TextCNN	0.79	0.79	0.72	0.75
BiGRU	0.78	0.75	0.75	0.75
BiLSTM	0.78	0.78	0.73	0.75
TF-IDF + LR	0.79	0.83	0.66	0.74
DistilBERT	0.81	0.83	0.75	0.79
ALBERT	0.79	0.82	0.71	0.76
MiniLM	0.82	0.80	0.80	0.80
MiniLM + LR	0.82	0.82	0.76	0.79
BiLSTM + Attention + MiniLM	0.83	0.89	0.69	0.78

VII. Discussion

Interpretation of Findings The results show that model performance improves as the ability to understand context increases. Traditional models such as TF-IDF with Logistic Regression performed reasonably well when tweets contained clear disaster-related keywords, but they often failed to correctly classify tweets written in informal language or containing slang and metaphors. Deep learning models including TextCNN, BiGRU, and BiLSTM showed better performance by learning word order and contextual relationships. Among them, BiLSTM achieved the best results because it can capture long-range dependencies and better handle ambiguous tweet content. Transformer-based models produced the highest overall performance due to their self-attention mechanism, which allows them to understand contextual meaning more effectively. Hybrid models that combine transformers with sequential models further improved performance while maintaining better interpretability.

A. Limitations

Despite strong results, this study has several limitations. Transformer-based models require higher computational resources, which can make real-time deployment difficult. The dataset used in this project contains only English tweets, limiting the model's ability to generalize to other languages. Additionally, performance may decrease when classifying tweets from different disaster types due to domain shift. The study also focused only on tweet text and did not use additional metadata such as location, hashtags, or timestamps, which could provide useful contextual information.

B. Improvements

Future improvements can address these limitations by using larger and multilingual datasets to improve generalization. Domain adaptation techniques can help models perform better across different disaster events. Incorporating tweet metadata and domain knowledge may further enhance classification accuracy. Additionally, few-shot and semi-supervised learning approaches can reduce dependence on large labeled datasets,

and model optimization techniques can improve efficiency for real-time disaster monitoring systems.

VIII. Conclusion

A. Summary

This project presented a comparative study of traditional machine learning, deep learning, transformer-based, and hybrid models for disaster tweet classification. The results clearly show that context-aware models outperform classical approaches. While TF-IDF-based models are simple and efficient, they struggle with informal and ambiguous tweet text. Deep learning models, especially BiLSTM, improved performance by capturing sequential information, while transformer-based and hybrid models achieved the best overall results by effectively modeling semantic and contextual relationships in tweets.

B. Future Work

Future work can focus on deploying these models in real-time disaster response systems. Extending the approach to multilingual and cross-domain datasets would improve usability in global disaster scenarios. Exploring few-shot and semi-supervised learning methods can help handle situations with limited labeled data, and improving model efficiency will support large-scale and real-world applications.

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